

[Secure Exit](#)

SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

Q1

Smart University Payroll
Management System
25 marks

**Q2**

Smart Vehicle Insurance Claim
System
25 marks



2/2 answered

Q2 Smart Vehicle Insurance Claim System 25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. **Requirements** Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount

Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks) **Problem Statement** An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. **Requirements** Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

$$\text{Claim} = \text{insuredAmount} \times 0.80$$

Bike

$$\text{Claim} = \text{insuredAmount} \times 0.60$$

Truck

$$\text{Claim} = \text{insuredAmount} \times 0.90$$

Test Case 1

Input

3

Car

TN10AB1234

Ramesh

800000

Bike

TN09XY5678

Suresh

100000

Truck

TN11TR9999

Logan

1500000

Search Vehicle:

TN09XY5678

Expected Output

Vehicle : TN10AB1234

Claim Amount : 640000.0

Vehicle : TN09XY5678

Claim Amount : 60000.0

Vehicle : TN11TR9999

Claim Amount : 1350000.0

Highest Claim Vehicle

TN11TR9999

1350000.0

Vehicle Found

Owner : Suresh

Claim Amount : 60000.0

Your Code

```
1 import java.util.*;
2
3 class Vehicle{
4     String no,owner;double amt;
5     Vehicle(String n,String o,double a){no=n;o=o;amt=a;}
6     double calculateClaimAmount(){return amt;}
7
8     public static void main(String[] args)
9     {
10         Scanner s = new Scanner(System.in);
11         int n=s.nextInt();
12         Vehicle[] v=new Vehicle[n];
13
14         for(int i=0;i<n;i++){
15             String t=s.next(),no=s.next(),
16             double a=s.nextDouble();
17
18             if(t.equalsIgnoreCase("Car"))
19             else if(t.equalsIgnoreCase("Bus"))
20             else v[i]=new Truck(no,o,a);
21         }
22         Vehicle high=v[0];
23         for(Vehicle x:v){
24             System.out.println("Vehicle : "+x.no);
25             System.out.println("Claim Amount : "+x.calculateClaimAmount());
26             System.out.println();
27             if(x.calculateClaimAmount()>high.calculateClaimAmount())
28                 high=x;
29         }
30         System.out.println("Highest Claim Amount : "+high.calculateClaimAmount());
31         System.out.println("Vehicle No : "+high.no);
32
33         String search=s.next();
34         boolean found = false;
35         for(Vehicle x:v){
36             if(x.no.equals(search)){
37                 System.out.println("Vehicle No : "+x.no);
38                 found = true;
39                 System.out.println("Owner : "+x.owner);
40                 System.out.println("Claim Amount : "+x.calculateClaimAmount());
41                 break;
42             }
43         }
44     }
45 }
```

```
42         }
43     }
44 }
45 class Car extends Vehicle{
46     Car(String n,String o,double a){
47         super(n,o,a);
48     }
49     double calculateClaimAmount(){
50         return amt*0.8;
51     }
52 }
53 class Bike extends Vehicle{
54     Bike(String n,String o,double a){
55         super(n,o,a);
56     }
57     double calculateClaimAmount(){
58         return amt*0.6;
59     }
60 }
61 class Truck extends Vehicle{
62     Truck(String n,String o,double a){
63         super(n,o,a);
64     }
65     double calculateClaimAmount(){
66         return amt*0.9;
67     }
68 }
```



Input

3

Car

TN10AB1234



Output

Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0



Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



0.00 / 25 marks

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